

School of Computer Science

A Level 1 Module COMP1038 Autumn 2025/26

Programming and Algorithms

Instructions:

Time allowed: One Hour

Answer ALL questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

No calculators are permitted in this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Exam Papers must not be removed from the examination venue.

Note: Answers

Knowledge classification: Following the Schools' recommendation, the (sub) questions have been classified as follows, using a subset of Bloom's Taxonomy:

K: Knowledge

C: Comprehension

A: Application

Question 1: Choose the correct answer to the following questions**[5 marks]**

- i. What will be the output of the following code snippet?

```
#include <stdio.h>
int main() {
    int a=5, b=2;
    printf("%d", a/b);
    return 0;
}
```

- a. 2.5
- b. 2
- c. 3
- d. Compilation error

[1 Mark]

- ii. Which of the following statements about the `static` keyword in C is correct?

- a. It limits the scope of a variable to the current file only.
- b. It limits the scope of a variable to the current function and maintains its value between function calls.
- c. It makes a function visible to other files.
- d. It allocates memory on the heap.

[1 Mark]

- iii. What is the output of the following code?

```
#include <stdio.h>
int main() {
    char s[] = "Hello";
    int i = 0;
    while(s[i]) {
        printf("%c", s[i++]);
    }
    return 0;
}
```

- a. H
- b. H e l l o
- c. Hello
- d. Compilation error

[1 Mark]

- iv. Given the following function, what will be printed when `func(5)` is called?

```
void func(int n) {
    if (n > 0) {
        func(n - 1);
        printf("%d", n);
    }
}
```

- a. 5 4 3 2 1
- b. 1 2 3 4 5
- c. 5
- d. 1 5

[1 Mark]

- v. Consider the following code for pointer arithmetic. What is the output?

```
#include <stdio.h>
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int *p = arr;
    printf("%d", *(p+3));
    return 0;
}
```

- a. 1
- b. 3
- c. 4
- d. 5

[1 Mark]

Answers:

- i. **[Application]** (Difficulty level: medium)

b [1 Mark]

- ii. **[Comprehension]** (Difficulty level: easy)

b [1 Mark]

- iii. **[Application]** (Difficulty level: medium)

c [1 Mark]

- iv. **[Application]** (Difficulty level: medium)

b [1 Mark]

- v. **[Application]** (Difficulty level: medium)

c [1 Mark]

Question 2: What is a NULL pointer in the C programming language? Is it equivalent to an uninitialized pointer? Provide code examples for both.

[5 Marks]

Answer:

[Knowledge, Comprehension, Application] (Difficulty level: easy)

A **NULL pointer** in C is a pointer that does not point to any valid memory address or object. The value of a NULL pointer is defined by the macro NULL (usually zero), indicating that the pointer is intentionally set to have no reference. This is useful for checking whether a pointer is pointing to something valid or not.

Example of a NULL pointer:

```
#include <stdio.h>

int main() {
    int *p = NULL; // p is a NULL pointer
    if (p == NULL) {
        printf("Pointer p is NULL and does not point to any memory location.\n");
    }
    return 0;
}
```

Is it equivalent to an uninitialized pointer?

No, a NULL pointer is **not** equivalent to an uninitialized pointer.

- **NULL pointer:** Is *explicitly* assigned the value NULL, indicating it points to nothing.
- **Uninitialized pointer:** Is *not assigned* any value, so it holds an unpredictable, garbage address (whatever happened to be in that memory location). Using an uninitialized pointer is dangerous and can cause bugs or crashes.

Example of an uninitialized pointer:

```
#include <stdio.h>

int main() {
    int *q; // q is an uninitialized pointer
    printf("%d", *q); // Dangerous! This may crash the program
    return 0;
}
```

Marking Scheme: 1 mark for defining the NULL pointer, 1 mark for its code example, 2 marks for writing their differences, and 1 mark for the code example of uninitialized pointer.

Question 3: Error Identification and Correction

The following C program is designed to count and print the number of even numbers in an array entered by the user. However, it contains several errors. Identify and correct these errors. Explain each correction that you made.

[5 Marks]

```
1  #include <stdio.h>

2  int main() {
3  int n, i, count;
4  int arr[10];
5  printf("Enter number of elements (not more than 10): ");
6  scanf("%d", n);

7  printf("Enter %d integers:\n", n);
8  for (i = 1; i <= n; i++) {
9      scanf("%d", arr[i]);
10 }

11 for (i = 0; i < n; i++);
12     if (arr[] % 2 == 0)
13         count++;

14 printf("Number of even numbers: %d", count);
15 return 0;
16 }
```

Answer:

[Comprehension, Application](Difficulty level: medium) Error Identification:

1. **Missing address operator in scanf for variable n in line 6:**
Should be `scanf("%d", &n);` not `scanf("%d", n);`
2. **Array input loop starts at index 1 and goes to n (inclusive) in line 8:**
Should be `for(i = 0; i < n; i++)` to cover valid indices 0 to n-1.
3. **Missing address operator in scanf for array input in line 9:**
Should be `scanf("%d", &arr[i]);` not `scanf("%d", arr[i]);`
4. **Uninitialized variable count before line 11:**
The count variable is not initialized, leading to garbage value. Should have `count = 0;` before using.
5. **Semicolon after the for loop causes an empty loop and incorrect even counting in line 11:**
The line `for(i = 0; i < n; i++);` has a semicolon at the end, making the subsequent `if` statement apply only once with `i==n`. Should remove the semicolon.
6. **Array index is missing in line 12:**
Should be `if(arr[i] % 2 == 0)` not `if(arr[] % 2 == 0)` to select individual elements from the Array.

Corrected Code:

```

1  #include <stdio.h>

2  int main() {
3      int n, i, count;
4      int arr[10];
5      printf("Enter number of elements: ");
6      scanf("%d", &n);

7      printf("Enter %d integers:\n", n);
8      for (i = 0; i < n; i++) {
9          scanf("%d", &arr[i]);
10     }

11     count = 0;
12     for (i = 0; i < n; i++)
13         if (arr[i] % 2 == 0)
14             count++;

15     printf("Number of even numbers: %d", count);
16     return 0;
17 }
```

Marking Scheme: 0.5 mark for identifying each error of any five errors in the program, and 0.5 mark for correcting each of the errors and providing explanations.

Question 4: Array Manipulation and Counting**[7 Marks]**

Problem Statement: Write a C program that reads an integer array of size n (where $2 \leq n \leq 20$) from the user, then determines and displays *all* unique pairs of elements whose sum is equal to a given target value (also input by the user).

For each pair, print the indices of the two elements (indices start at 0) and their values. If no such pairs are found, print an appropriate message.

Answer:**[Comprehension, Application] (Difficulty level: hard)**

```
#include <stdio.h>

int main() {
    int n, arr[20], target, i, j, found = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);
    if (n < 2 || n > 20) {
        printf("Invalid size. Please enter n between 2 and 20.\n");
        return 1;
    }

    printf("Enter array elements: ");
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter target sum: ");
    scanf("%d", &target);

    printf("\nPairs with sum %d:\n", target);
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] + arr[j] == target) {
                printf("Indices: %d, %d => Values: %d, %d\n", i, j, arr[i], arr[j]);
                found = 1;
            }
        }
    }
    if (!found)
        printf("No pairs found with the given sum.\n");

    return 0;
}
```

Marking Scheme: correct input & validation – 2 marks, nested loop logic for unique pairs – 2 marks, correct sum check – 1 mark, clear output (pairs or message) – 2 marks

Question 5: Explain with an example how a nested structure (structure within a structure) can be defined and used in C to represent hierarchical data.

[3 Marks]

Answer:**[Knowledge](Difficulty level: easy)**

In C, nested structures allow hierarchical organization of data by including one structure as a member of another. This is useful for representing entities that have sub-attributes grouped together.

Example:

```
struct Date {
    int day;
    int month;
    int year;
};

struct Employee {
    char name[50];
    int id;
    struct Date joining_date; // Nested structure
};
```

Here, the Employee structure contains a Date structure for the joining date. This approach makes managing complex, related data more organized and efficient.

Marking Scheme: 1 mark for explaining the nested structure, 1 mark for defining the inner structure, and 1 mark for defining the outer structure.

--- End ---